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ELECTRODE ARRAY FOR SPATIALLY RANDOM ELECTRICAL STIMULATION

Abstract

An electrical stimulation system for delivering spatially random electrical stimulation to a patient through an electrode array according to one embodiment includes an electrode array comprising a plurality of electrodes spaced apart from one another, a signal generator electrically coupled to the electrode array and configured to deliver an electrical stimulation signal, and a controller comprising a processor and a memory having a plurality of instructions stored thereon that, in response to execution by the processor, causes the controller to randomly select an electrical configuration from the plurality of electrodes of the electrode array by randomly selecting a first set of electrodes of the plurality of electrodes to function as electrical sources and a second set of electrodes of the plurality of electrodes to function as electrical sinks, and to instruct the signal generator to deliver the electrical stimulation signal to the patient via the randomly selected electrical configuration.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 17/984,062, filed on Nov. 9, 2022, the contents of which are incorporated herein by reference in their entirety.

BACKGROUND

[0002] Electrical stimulation technologies are used to mitigate pain and other conditions. The technologies work by delivering electrical stimulation to the various parts of the nervous system (e.g., sensory receptors, peripheral nerves, spinal cord, and brain). For example, the stimulation devices used today deliver pulsed periodic waveforms through electrical contacts that are positioned overtop of the dorsal columns of the spinal cord, dorsal root ganglion, or long peripherals nerves. The electrical contacts that are used for stimulation are chosen at the onset of the treatment and maintained throughout the duration of the therapy, which may last for years. Although the technologies used today cause a moderate reduction in pain intensity in the majority of the persons treated, the effects of the stimulation often decline after a few years of use. This phenomenon is known as neurological tolerance (or accommodation), and it is the leading cause of stimulator removal. Electrical stimulation technologies are used to treat a variety of disease states, such as chronic pain, overactive bladder, movement disorders, and cardiovascular disease. However, the treatment effects and time-course are limited.

SUMMARY

[0003] One embodiment is directed to a unique system and method for delivering spatially random electrical stimulation to a patient through an electrode array. Other embodiments are directed to apparatuses, systems, devices, hardware, methods, and combinations thereof for delivering spatially random electrical stimulation to a patient through an electrode array.

[0004] According to an embodiment, an electrical stimulation system for delivering spatially random electrical stimulation to a patient through an electrode array may include an electrode array comprising a plurality of electrodes spaced apart from one another, a signal generator electrically coupled to the electrode array and configured to deliver an electrical stimulation signal, and a controller comprising a processor and a memory having a plurality of instructions stored thereon that, in response to execution by the processor, causes the controller to randomly select an electrical configuration from the plurality of electrodes of the electrode array, wherein to randomly select the electrical configuration includes to randomly select a first set of electrodes of the plurality of electrodes to function as electrical sources and a second set of electrodes of the plurality of electrodes, different from the first set of electrodes, to function as electrical sinks, and instruct the signal generator to deliver the electrical stimulation signal to the patient via the randomly selected electrical configuration.

[0005] In some embodiments, the first set of electrodes may include a first number of electrodes and the second set of electrodes may include a second number of electrodes different from the first number.

[0006] In some embodiments, the plurality of electrodes may include a third set of electrodes of a first cross-sectional shape and a fourth set of electrodes of a second cross-sectional shape different from the first cross-sectional shape.

[0007] In some embodiments, the first cross-sectional shape may be a circular shape.

[0008] In some embodiments, the plurality of electrodes may include a third set of electrodes of a first size and a fourth set of electrodes of a second size greater than the first size.

[0009] In some embodiments, the signal generator may be configured to deliver low-frequency signal components through only the fourth set of electrodes.

[0010] In some embodiments, the plurality of electrodes may include a third set of electrodes and a fourth set of electrodes different from the third set of electrodes, and to instruct the signal generator to deliver the electrical stimulation signal to the patient via the randomly selected electrical configuration may include to instruct the signal generator to deliver a first set of features of the electrical stimulation signal to the patient via the third set of electrodes and a second set of features of the electrical stimulation signal, different from the first set of features, to the patient via the fourth set of electrodes.

[0011] In some embodiments, the first set of features may include one or more of a frequency range of the electrical stimulation signal or an amplitude range of the electrical stimulation signal.

[0012] In some embodiments, the electrical stimulation system may further include an electrode assembly that includes the plurality of electrodes, the third set of electrodes may be positioned on the electrode assembly at a first section of the electrode assembly, and the fourth set of electrodes may be positioned on the electrode assembly at a second section of the electrode assembly different from the first section.

[0013] In some embodiments, each of the third set of electrodes and the fourth set of electrodes may be selected based on one or more of an electrode size, an electrode material, or an electrode location on an electrode assembly.

[0014] In some embodiments, the plurality of electrodes may include a third set of electrodes composed of a first material and a fourth set of electrodes composed of a second material different from the first material.

[0015] In some embodiments, the plurality of instructions may further cause the controller to randomly assign a polarity to one or more of the electrodes of the first set of electrodes and the second set of electrodes.

[0016] In some embodiments, the electrode array may include at least a distal section and a proximal section, and the signal generator may be configured to deliver high-frequency signal components through only the distal section of the electrode array.

[0017] In some embodiments, the electrode array may include at least a distal section and a proximal section, and the signal generator may be configured to amplify signals transmitted through the distal section at a different weight from signals transmitted through the proximal section.

[0018] In some embodiments, the electrical stimulation signal may include or be embodied as a noise signal.

[0019] In some embodiments, the plurality of instructions may further cause the controller to partition a frequency range of the electrical stimulation signal into a plurality of discrete frequency bands, each of the discrete frequency bands having a corresponding bandwidth, adjust an amplitude of one or more of a voltage or a current of the electrical stimulation signal within a selected frequency band of the plurality of discrete frequency bands to generate an adjusted electrical signal based on feedback received from the patient, and instruct the signal generator to deliver the adjusted electrical signal to the patient via the randomly selected electrical configuration.

[0020] In some embodiments, the feedback may include at least one of feedback received from the patient as patient self-report regarding the therapy delivered to the patient, feedback based on data generated by one or more sensors of the electrical stimulation system, wherein the one or more sensors measure one or more physiological outcomes of the patient, or feedback data received from a machine learning system.

[0021] In some embodiments, the frequency range of the electrical stimulation signal may be 0.05 Hz to 2 kHz.

[0022] In some embodiments, the frequency range of the electrical stimulation signal may be 1 kHz to 100 kHz.

[0023] In some embodiments, the frequency range of the electrical stimulation signal may be 100 kHz to 1 MHz.

[0024] In some embodiments, the plurality of instructions may further cause the controller to randomly select a new electrical configuration from the plurality of electrodes of the electrode array in response to a determination that a threshold has been met.

[0025] In some embodiments, the threshold may be defined as a predefined period of time elapsed since randomly selecting the electrical configuration.

[0026] In some embodiments, the threshold may be associated with one or more characteristics of the spatially random electrical stimulation being delivered to the patient.

[0027] According to another embodiment, a method of delivering spatially random electrical stimulation to a patient through an electrode array that includes a plurality of electrodes may include randomly selecting, by an electrical stimulation system, an electrical configuration from the plurality of electrodes of the electrode array, wherein randomly selecting the electrical configuration includes randomly selecting a first set of electrodes of the plurality of electrodes to function as electrical sources and a second set of electrodes of the plurality of electrodes, different from the first set of electrodes, to function as electrical sinks, and delivering, by the electrical stimulation system, an electrical stimulation signal to the patient via the randomly selected electrical configuration.

[0028] In some embodiments, the method may further include partitioning, by the electrical stimulation system, a frequency range of the electrical stimulation signal into a plurality of discrete frequency bands, each of the discrete frequency bands having a corresponding bandwidth, and adjusting, by the electrical stimulation system, an amplitude of one or more of a voltage or a current of the electrical stimulation signal within a selected frequency band of the plurality of discrete frequency bands to generate an adjusted electrical signal based on feedback received from the patient, wherein delivering the electrical stimulation signal to the patient includes delivering the adjusted electrical signal to the patient via the randomly selected electrical configuration.

[0029] In some embodiments, the feedback may include at least one of feedback received from the patient as patient self-report regarding the therapy delivered to the patient, feedback based on data generated by one or more sensors of the electrical stimulation system, wherein the one or more sensors measure one or more physiological outcomes of the patient, or feedback data received from a machine learning system.

[0030] In some embodiments, the first set of electrodes may include a first number of electrodes and the second set of electrodes may include a second number of electrodes different from the first number.

[0031] In some embodiments, the plurality of electrodes may include a third set of electrodes of a first cross-sectional shape and a fourth set of electrodes of a second cross-sectional shape different from the first cross-sectional shape.

[0032] In some embodiments, the plurality of electrodes may include a third set of electrodes of a first size and a fourth set of electrodes of a second size greater than the first size.

[0033] In some embodiments, the plurality of electrodes may include a third set of electrodes composed of a first material and a fourth set of electrodes composed of a second material different from the first material.

[0034] This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter. Further embodiments, forms, features, and aspects of the present application shall become apparent from the description and figures provided herewith.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0035] The concepts described herein are illustrative by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, elements illustrated in the figures are not necessarily drawn to scale. Where considered appropriate, references labels have been repeated among the figures to indicate corresponding or analogous elements.

[0036] FIG. 1 illustrates at least one embodiment of a system for delivering one or more electrical signals to target neural tissue, non-neural tissue, or a combination thereof in a patient, where the target tissue is located within or adjacent to the patient's brain.

[0037] FIG. 2 illustrates at least one embodiment of a system for delivering one or more electrical signals to target neural tissue, non-neural tissue, or a combination thereof in a patient, where the target tissue is located within or adjacent the spinal cord;

[0038] FIG. 3 is a zoomed-in view of the spinal cord and illustrates at least one option for electrode placement according to the system of FIG. 2, where the target tissue is located within or adjacent a dorsal region of the spinal cord, such as the dorsal columns;

[0039] FIG. 4 is a zoomed-in view of the spinal cord and illustrates at least one other option for electrode placement according to the system of FIG. 2, where the target tissue is located within or adjacent to the dorsolateral region of the spinal cord, such as the dorsolateral funiculus.

[0040] FIG. 5 is a zoomed-in view of the spinal cord and illustrates at least one other option for electrode placement according to the system of FIG. 2, where the target tissue is located within or adjacent the lateral region of the spinal cord, such as the spinothalamic tract;

[0041] FIG. 6 is a zoomed-in view of the spinal cord and illustrates at least one other option for electrode placement according to the system of FIG. 2, where the target tissue is located with or adjacent the ventral region of the spinal cord, such as the ventral horn;

[0042] FIG. 7 is a zoomed-in view of the spinal cord and illustrates at least one other option for electrode placement according to the system of FIG. 2, where the target tissue is located within or adjacent a dorsal root ganglion;

[0043] FIG. 8 illustrates at least one embodiment of a system for delivering one or more electrical signals to target neural tissue, non-neural tissue, or a combination thereof in a patient, where the target tissue is located within or adjacent a sympathetic chain ganglion;

[0044] FIG. 9 is a zoomed-in view of the sympathetic chain and illustrates at least one option for electrode placement according to the system of FIG. 8;

[0045] FIG. 10 illustrates at least one embodiment of a system for delivering one or more electrical signals to target neural tissue, non-neural tissue, or a combination thereof in a patient, where the target tissue is located within or adjacent a peripheral nerve;

[0046] FIG. 11 is a simplified block diagram of at least one embodiment of an electrical stimulation system for providing therapy to a patient via the application of one or more electrical signals;

[0047] FIG. 12 is a simplified block diagram of at least one embodiment of a computing system;

[0048] FIG. 13 illustrates at least one embodiment of an electrode array of an electrical stimulation device;

[0049] FIGS. 14-17 illustrates various electrical configurations of the electrode array of FIG. 13;

[0050] FIG. 18 illustrates at least one other embodiment of an electrode array of an electrical stimulation device;

[0051] FIG. 19 illustrates one or more electrical configuration of the electrode array of FIG. 18;

[0052] FIG. 20 illustrates at least one other embodiment of an electrode array of an electrical stimulation device and one or more electrical configurations thereof;

[0053] FIG. 21 is a simplified flow diagram of at least one embodiment of a method of randomly

selecting an electrical configuration of an electrode array for delivering electrical stimulation to a patient;

[0054] FIG. 22 is a simplified flow diagram of at least one embodiment of a method of providing spatially random electrical stimulation to a patient;

[0055] FIG. 23 is a graph that illustrates partitioned frequency bands of at least a portion of a frequency range of an electrical signal; and

[0056] FIG. 24 is a graph of the power spectrum of a broad spectrum electrical noise signal that has been partitioned into discrete frequency bands.

DETAILED DESCRIPTION

[0057] Although the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described herein in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.

[0058] References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. It should further be appreciated that although reference to a “preferred” component or feature may indicate the desirability of a particular component or feature with respect to an embodiment, the disclosure is not so limiting with respect to other embodiments, which may omit such a component or feature. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to implement such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. Additionally, it should be appreciated that items included in a list in the form of “at least one of A, B, and C” can mean (A); (B); (C); (A and B); (B and C); (A and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (B and C); (A and C); or (A, B, and C). Further, with respect to the claims, the use of words and phrases such as “a,” “an,” “at least one,” and/or “at least one portion” should not be interpreted so as to be limiting to only one such element unless specifically stated to the contrary, and the use of phrases such as “at least a portion” and/or “a portion” should be interpreted as encompassing both embodiments including only a portion of such element and embodiments including the entirety of such element unless specifically stated to the contrary.

[0059] The disclosed embodiments may, in some cases, be implemented in hardware, firmware, software, or a combination thereof. The disclosed embodiments may also be implemented as instructions carried by or stored on one or more transitory or non-transitory machine-readable (e.g., computer-readable) storage media, which may be read and executed by one or more processors. A machine-readable storage medium may be embodied as any storage device, mechanism, or other physical structure for storing or transmitting information in a form readable by a machine (e.g., a volatile or non-volatile memory, a media disc, or other media device).

[0060] In the drawings, some structural or method features may be shown in specific arrangements and/or orderings. However, it should be appreciated that such specific arrangements and/or orderings may not be required. Rather, in some embodiments, such features may be arranged in a different manner and/or order than shown in the illustrative figures unless indicated to the contrary. Additionally, the inclusion of a structural or method feature in a particular figure is not meant to imply that such feature is required in all embodiments and, in some embodiments, may not be included or may be combined with other features.

[0061] As described above, existing electrical stimulation technologies are inhibited from long-term efficacy due to the patient often developing neurological tolerance (or accommodation) to the

stimulation. It should be appreciated that the technologies described herein include an electrical stimulation system that delivers electrical stimuli in a random fashion through a set of electrodes (e.g., electrical contacts) to neural and/or non-neural tissue, which has been found to be more effective at treating acute and chronic pain and other medical disorders than existing technologies. For example, in illustrative embodiments, the electrical stimulation system includes an electrode array that delivers spatially random electrical stimulation to the patient, which can avoid delivering a repetitive barrage of electrical stimulation to the same biological tissue and thereby avoid eliciting neurological tolerance in the patient.

[0062] As described in greater detail below, the therapy may be generated by an external or implantable electrical stimulator and delivered through electrodes (e.g., an electrode array) to the patient's brain or spinal cord, a dorsal root ganglion, a sympathetic nerve or chain ganglion, a cranial nerve, a parasympathetic nerve, or a peripheral nerve. The therapy may be used to treat pain (e.g., chronic pain), an autonomic disorder (e.g., diabetic peripheral neuropathy, hypertension, hypotension, complex regional pain syndrome (CRPS), Raynaud's syndrome, overactive bladder, urinary incontinence, fecal incontinence, fecal constipation, migraine, etc.), a sensory disorder (e.g., tinnitus, hearing loss, vertigo, etc.), a motor disorder (e.g., Huntington's disease, Parkinson's disease, Multiple Sclerosis, spinal muscular atrophy (SMA), dystonia, essential tremor, etc.), or a combination thereof. Further, the therapy provided to the patient can elicit plastic changes in neural tissue, non-neural tissue, or a combination thereof to mitigate or abolish a pathophysiologic disease or syndrome. Plastic changes are changes to the neural tissue, non-neural tissue, or a combination thereof in response to physiological demands. Such plastic changes can include morphological and functional changes.

[0063] Whether the electrical signal(s) are being applied to target neural tissue, non-neural tissue, or a combination thereof located within or adjacent the patient's brain or spinal cord, a dorsal root ganglion, a sympathetic chain ganglion, a cranial nerve, autonomic circuitry, or a peripheral nerve, the technologies described herein illustrate that the specific electrical configuration of the electrode array, the specific parameters of the electrical signals, and/or the location of the electrode array through which electrical signals are delivered can be selectively controlled to provide improved symptom relief and therapy to the patient for the treatment of pain, autonomic disorders, sensory disorders, motor disorders, etc. Additionally, the specific physical design of the electrode array may be selected. The specific system and parameters are discussed in more detail below.

[0064] Referring now to FIG. 1, there is illustrated a system 50 for delivering one or more electrical signals to provide therapy to a patient 116, where the target neural tissue, non-neural tissue, or a combination thereof 135 is located within or adjacent tissue within the patient's brain 134. In general, the system 50 in FIG. 1 can include one or more electrodes 107 (shown diagrammatically in FIG. 1) that are connected by an electrical lead 108 to a signal generator 109. As described herein, multiple electrodes 107 may be arranged into an electrode array in some embodiments (see, e.g., FIGS. 11-18). In various embodiments, one or more electrodes 107 (or electrical contacts) may be embodied on, form a portion of, or be electrically coupled to one or more electrical leads 108 that are electrically (or electromagnetically) coupled to the signal generator 109. An additional lead 117 can be used to couple the signal generator 109 to the rest of the system 50, which can include a user interface 112 and a controller 110, where it is to be understood that as an alternative to the use of the lead 117, the signal generator 109 can be wirelessly connected to the rest of the system 50. The system 50 can also include a power system 111 and/or a patient monitor system. Further, it should be understood that while the system 50 of FIG. 1 illustrates a configuration where electrical signals can be delivered to target neural tissue, non-neural tissue, or a combination 135 thereof utilizing one or more electrodes 107 (e.g., an electrode array) coupled to an implantable signal generator 109 via a lead 108, the electrode(s) 107 can alternatively be coupled to an external signal generator via a wireless antenna system. It should be appreciated that, in some embodiments, more than one electrode 107, more than one electrical

lead **108**, and/or more than one signal generator **109** may be used in the system **50**. Regardless of the exact type (e.g., percutaneous, transcutaneous, implantable, etc.) or configuration (e.g., monopolar, bipolar, multipolar, etc.) of the electrode(s) **107**, the electrode(s) **107** can be in the form of an electrode assembly that can deliver electrical signals to a patient to provide therapy to the patient and/or improve one or more of the patient's symptoms. Specific diseases or conditions that can be treated based on stimulation of the brain include, for example, Parkinson's disease, essential tremor, depression, obsessive compulsive disorder, Tourette's syndrome, epilepsy, schizophrenia, narcolepsy, seizures, Alzheimer's disease, tinnitus, Meniere's disease, and chronic pain.

[0065] Referring now to FIG. **2**, there is illustrated a system **100** for delivering one or more electrical signals to provide therapy to a patient, where the target neural tissue, non-neural tissue, or a combination thereof is located within or adjacent the spinal cord **101** of a patient **116**. As shown in FIG. **2**, the system **100** can include multiple devices to control and deliver electrical signals to one or more areas of target neural tissue, non-neural tissue, or a combination thereof located within or adjacent the spinal cord **101** to provide therapy to a patient **116**. In general, the system **100** in FIG. **2** can include one or more electrodes **107a** and/or **107b** that are connected by one or more electrical leads **108** to a signal generator **109**. As described herein, multiple electrodes **107** may be arranged into an electrode array in some embodiments (see, e.g., FIGS. **11-18**). An additional lead **117** can be used to couple the signal generator **109** to other components of the system **100**, which can include a user interface **112** and a controller **110**, where it is to be understood that as an alternative to the use of the lead **117**, the signal generator **109** can be wirelessly connected to the rest of the system **50**. The system can also include an isolated power system **111** and/or a patient monitor system. Further, it should be understood that while the system **100** of FIG. **2** illustrates a configuration where electrical signals can be delivered to target neural tissue, non-neural tissue, or a combination thereof utilizing electrodes **107a** and/or **107b** coupled to an implantable signal generator **109** via a lead **108**, the electrodes **107a** and/or **107b** can alternatively be coupled to an external signal generator via a wireless antenna system. Regardless, the electrodes **107a** and/or **107b** can be in the form of an electrode assembly that can deliver electrical signals to a patient to provide therapy to the patient and/or improve one or more of the patient's symptoms based on, for example, the specific location of the electrodes, as discussed in more detail in FIGS. **3-7** below.

[0066] Referring now to FIG. **3**, the placement of the electrode or electrodes **107** (e.g., electrode array) in order to deliver one or more electrical signals to an area within or adjacent target neural tissue, non-neural tissue, or a combination thereof **131** located adjacent a dorsal region **120** of the spinal cord **101**, and in particular a dorsal column **118**, is discussed in more detail, where the dorsal D and ventral V directions of the spinal cord **101** are labeled for reference purposes. For instance, one or more electrodes **107** can be positioned within a portion of the epidural space **128** of the patient **116** adjacent a dorsal region **120** of the spinal cord **101**, where the dorsal region **120** of the spinal cord **101** can be identified via locating the posterior median sulcus **125**. As shown, the epidural space **128** is positioned between the bone **129** (vertebrae) and the dura mater **115**. Thus, electrical signals transmitted by the electrode(s) **107** must be configured to pass through the dura mater **115**, subdural cavity **113**, arachnoid mater **114**, subarachnoid cavity **130**, and pia mater **127** to reach the target neural tissue, non-neural tissue, or a combination thereof **131** and deliver the desired electrical signals therein. By placing the electrode or electrodes **107** in the epidural space **128** adjacent a dorsal region **120** of the spinal cord **101**, electrical signals can be delivered to target neural tissue, non-neural tissue, or a combination thereof **131** located within or adjacent a dorsal column **118** to provide therapy to the patient. It is also to be understood that the electrode or electrodes **107** can be positioned in any suitable location in the dorsal region **120** of the spinal cord **101** in order to deliver electrical signals to an area within or adjacent other target neural tissue, non-neural tissue, or a combination thereof, such as tissue located adjacent a dorsal horn **119** or a dorsal root **121**. Specific diseases or conditions that can be treated based on stimulation of the dorsal region of the spinal cord, and in particular, the dorsal columns include, for example, acute pain,

failed back surgery syndrome, complex regional pain syndrome, peripheral vascular disease and chronic limb ischemia, angina pain, diabetic pain, abdominal/visceral pain syndrome, brachial plexitis, phantom limb pain, intractable pain secondary to spinal cord injury, mediastinal pain, Raynaud's syndrome, cervical neuritis, post herpetic neuralgia, vertigo, tinnitus, hearing loss and inflammatory pain such as arthritis, irritable bowel pain, osteoarthritis pain, and fibromyalgia.

[0067] Referring now to FIG. 4, the placement of the electrode or electrodes **107** (e.g., an electrode array) in order to deliver one or more electrical signals to an area within or adjacent target neural tissue, non-neural tissue, or a combination thereof **137** located in a dorsolateral region **138** of the spinal cord **101** is discussed in more detail, where the dorsal D, ventral V, and lateral L directions are labeled for reference purposes. For instance, one or more electrodes **107** can be positioned adjacent a dorsolateral region **138** of the spinal cord **101**. By placing the electrode or electrodes **107** adjacent a dorsolateral region **138** of the spinal cord **101**, electrical signals can be delivered to target neural tissue, non-neural tissue, or a combination thereof **137** located within or adjacent a dorsolateral region **138** of the spinal cord **101** to provide therapy to the patient. Specifically, nerve fiber activity in the right or left dorsolateral funiculus **136** or a combination thereof can be altered via electrical signals in order to treat or alleviate symptoms associated various conditions. Specific diseases or conditions that can be treated based on stimulation of the dorsolateral region of the spinal cord, and in particular, the dorsolateral funiculus include, for example, acute pain, failed back surgery syndrome, complex regional pain syndrome, peripheral vascular disease and chronic limb ischemia, angina pain, diabetic pain, abdominal/visceral pain syndrome, brachial plexitis, phantom limb pain, intractable pain secondary to spinal cord injury, mediastinal pain, Raynaud's syndrome, cervical neuritis, post herpetic neuralgia, vertigo, tinnitus, hearing loss and inflammatory pain such as arthritis, irritable bowel pain, osteoarthritis pain, and fibromyalgia.

[0068] Referring now to FIG. 5, the placement of the electrode or electrodes **107** (e.g., an electrode array) in order to deliver one or more electrical signals to an area within or adjacent target neural tissue, non-neural tissue, or a combination thereof **218** located in a lateral region **219** of the spinal cord **101** is discussed in more detail, where the dorsal D, ventral V, and lateral L directions are labeled for reference purposes. For instance, one or more electrodes **107** can be positioned adjacent a lateral region **219** of the spinal cord **101**. By placing the electrode or electrodes **107** adjacent lateral region **219** of the spinal cord **101**, electrical signals can be delivered to target neural tissue, non-neural tissue, or a combination thereof **218** located within or adjacent a lateral region **219** of the spinal cord **101** to provide therapy to the patient. Specifically, nerve fiber activity in the right lateral spinothalamic tract **102**, the left lateral spinothalamic tract **104**, or a combination thereof can be altered via electrical signals in order to treat or alleviate symptoms associated various conditions. Moreover, it is to be understood that nerve fiber activity in the right anterior spinothalamic tract **103**, the left anterior spinothalamic tract **105**, or a combination thereof can also be altered via electrical signals based on the specific positioning of the one or more electrodes **107**. Specific diseases or conditions that can be treated based on stimulation of the lateral region of the spinal cord, and in particular, the lateral spinothalamic tract include, for example, acute pain, failed back surgery syndrome, complex regional pain syndrome, peripheral vascular disease and chronic limb ischemia, angina pain, diabetic pain, abdominal/visceral pain syndrome, brachial plexitis, phantom limb pain, intractable pain secondary to spinal cord injury, mediastinal pain, Raynaud's syndrome, cervical neuritis, post herpetic neuralgia, vertigo, tinnitus, hearing loss and inflammatory pain such as arthritis, irritable bowel pain, osteoarthritis pain, and fibromyalgia.

[0069] Referring now to FIG. 6, the placement of the electrode or electrodes **107** (e.g., an electrode array) in order to deliver one or more electrical signals to an area within or adjacent target neural tissue, non-neural tissue, or a combination thereof **147** located in a ventral region **220** of the spinal cord **101** is discussed in more detail, where the dorsal D and ventral V directions are labeled for reference purposes. For instance, one or more electrodes **107** can be positioned within a portion of the epidural space **128** of the patient **116** adjacent a ventral region **220** of the spinal cord **101**,

where the ventral region **220** of the spinal cord **101** can be identified via locating the anterior median fissure **106**. As shown, the epidural space **128** is positioned between the bone **129** (vertebrae) and the dura mater **115**. Thus, electrical signals transmitted by the electrode(s) **107** are configured to pass through the dura mater **115**, subdural cavity **113**, arachnoid mater **114**, subarachnoid cavity **130**, and pia mater **127** to reach the target neural tissue, non-neural tissue, or a combination thereof **147** and deliver the desired electrical signals therein. By placing the electrode or electrodes **107** in the epidural space **128** adjacent a ventral region **220** of the spinal cord **101**, electrical signals can be delivered to target neural tissue, non-neural tissue, or a combination thereof **147** located in a ventral region **220** of the spinal cord **101** to provide therapy to the patient. Specifically, in one particular embodiment, nerve fiber activity in the right or left ventral horn **146** or a combination thereof can be altered via electrical signals in order to treat or alleviate symptoms associated various conditions. Specific diseases or conditions that can be treated based on stimulation of the ventral region of the spinal cord include, for example, motoneuron disease (amyotrophic lateral sclerosis; progressive muscular atrophy; progressive bulbar palsy; primary lateral sclerosis; hereditary spastic paraplegia), spinal muscular atrophy (infantile and juvenile spinal muscular atrophy; focal amyotrophy), and multiple sclerosis.

[0070] Referring now to FIG. 7, the placement of the electrode or electrodes **107** (e.g., an electrode array) in order to deliver one or more electrical signals to an area within or adjacent target neural tissue, non-neural tissue, or a combination thereof **124** located adjacent or near a dorsal region **120** of the spinal cord **101**, and in particular a dorsal root ganglion **122**, is discussed in more detail, where the dorsal D and ventral V directions of the spinal cord **101** are labeled for reference purposes. For instance, one or more electrodes **107** can be positioned within a portion of the epidural space **128** of the patient **116** adjacent a dorsal (or posterior) portion **120** of the spinal cord **101**, where the dorsal (or posterior) portion **120** of the spinal cord **101** can be identified via locating the posterior median sulcus **125**. As shown, the epidural space **128** is positioned between the bone **129** (vertebrae) and the dura mater **115**. Thus, electrical signals transmitted by the electrode(s) **107** are configured to pass through the dura mater **115**, subdural cavity **113**, arachnoid mater **114**, subarachnoid cavity **130**, and pia mater **127** to reach the target neural tissue, non-neural tissue, or a combination thereof **124** and deliver the desired electrical signals therein. By placing the electrode or electrodes **107** in the epidural space **128** adjacent a dorsal D (or posterior) portion **120** of the spinal cord **101**, electrical signals can be delivered to target neural tissue, non-neural tissue, or a combination thereof **124** located within or adjacent a dorsal root ganglion **122** to provide therapy to the patient. Specific diseases or conditions that can be treated based on stimulation of the dorsal root ganglion include, for example, acute pain, failed back surgery syndrome, complex regional pain syndrome, peripheral vascular disease and chronic limb ischemia, angina pain, diabetic pain, abdominal/visceral pain syndrome, brachial plexitis, phantom limb pain, intractable pain secondary to spinal cord injury, mediastinal pain, Raynaud's syndrome, cervical neuritis, post herpetic neuralgia, vertigo, tinnitus, hearing loss and inflammatory pain such as arthritis, irritable bowel pain, osteoarthritis pain and fibromyalgia.

[0071] Referring now to FIG. 8, there is illustrated a system **300** for delivering one or more electrical signals to provide therapy to a patient **116**, where the target neural tissue, non-neural tissue, or a combination thereof **318** located adjacent a ventral or anterior region **219** of a spinal cord **101** of the patient **116**. It should be appreciated that the system **300** of FIG. 8 may include similar elements and/or features to the system **50** described in reference to FIG. 1. In particular, the target neural tissue, non-neural tissue, or a combination thereof **318** can be a sympathetic chain ganglion located in the right sympathetic chain **201**, the left sympathetic chain **202**, or a combination thereof. As shown in FIG. 8, in some embodiments, the system **300** can include multiple devices to control and deliver electrical signals to an area within or adjacent target neural tissue, non-neural tissue, or a combination thereof **318** located adjacent a ventral V (or anterior) region **219** of the spinal cord **101** to provide therapy to the patient **116**. In general, the system **300**

of FIG. 8 can include one or more electrodes **107** that are connected by an electrical lead **108** to a signal generator **109**. As described herein, multiple electrodes **107** may be arranged into an electrode array in some embodiments (see, e.g., FIGS. **11-18**). An additional lead **117** can be used to couple the signal generator **109** to the rest of the system **300**, which can include a user interface **112**, and a controller **110**, where it is to be understood that as an alternative to the use of the lead **117**, the signal generator **109** can be wirelessly connected to the rest of the system **50**. The system may also include an isolated power system **111** and/or a patient monitor system. Further, it should be understood that while the system **300** of FIG. 8 illustrates a configuration where electrical signals can be delivered to target neural tissue, non-neural tissue, or a combination thereof utilizing an electrode or electrodes **107** coupled to an implantable signal generator **109** via a lead **108**, the electrode or electrodes **107** can alternatively be coupled to an external signal generator via a wireless antenna system. Regardless, the electrode or electrodes **107** can be in the form of an electrode assembly that can deliver electrical signals to a patient to provide therapy to the patient and/or improve one or more of the patient's symptoms.

[0072] Referring now to FIG. 9, the placement of the electrode or electrodes **107** (e.g., an electrode array) is discussed in more detail. For instance, one or more electrodes **107** can be positioned adjacent a region of the right sympathetic chain **201** or the left sympathetic chain **202** of the patient **116**, where the sympathetic chains **201** and **202** are located ventral and lateral to a ventral (or anterior) region **219** of the spinal cord **101**. By placing the electrode or electrodes **107** adjacent target neural tissue, non-neural tissue, or a combination thereof **318** located lateral and ventral to a ventral (or anterior) region **219** of the spinal cord **101**, one or more electrical signals can be delivered to the target neural tissue, non-neural tissue, or a combination thereof **318** (e.g., a ganglion or ganglia of the right sympathetic chain **201** or the left sympathetic chain **202**) to provide therapy to the patient.

[0073] For instance, electrical signals can be delivered to a ganglion or ganglia associated with the cervical portion **203**, the thoracic portion **204**, the lumbar portion **205**, or the sacral portion **206** of the right sympathetic chain **201** or the left sympathetic chain **202**, or any combination thereof to provide therapy to the targeted area or areas. In one particular embodiment, one or more electrodes **107** (e.g., an electrode array) can be placed adjacent the cervical region **203** of the sympathetic chain to affect nerve fiber activity associated with levels C1-C3, which can affect nerve fiber activity associated with the eyes, the lachrymal glands, the salivary glands, and the sweat glands, hair follicles, and blood vessels of the head, neck, and arms. In another embodiment, one or more electrodes **107** (e.g., an electrode array) can be placed adjacent levels T1-T4 of the thoracic region **204**, which can affect nerve fiber activity associated with the heart and lungs. In an additional embodiment, one or more electrodes **107** (e.g., an electrode array) can be placed adjacent levels T5-T9 of the thoracic region **204**, which can affect nerve fiber activity associated with the stomach, duodenum, pancreas, liver, kidneys, and adrenal medulla. In yet another embodiment, one or more electrodes **107** (e.g., an electrode array) can be placed adjacent levels T10-T11 of the thoracic region **204**, which can affect nerve fiber activity associated with the stomach and duodenum. In one more embodiment, one or more electrodes **107** (e.g., an electrode array) can be placed adjacent level T12 of the thoracic region **204** and levels L1-L3 of the lumbar region **205**, which can affect nerve fiber activity in the colon, rectum, bladder, and external genitalia. In still another embodiment, one or more electrodes **107** (e.g., an electrode array) can be placed adjacent levels L4-L5 of the lumbar region **205** and levels S1-S3 of the sacral region **206**, which can affect nerve fiber activity associated with the sweat glands, hair follicles, and blood vessels of the lower limbs. In another embodiment, one or more electrodes **107** (e.g., an electrode array) can be placed adjacent levels S4-S5 of the sacral region **206**, which can affect nerve fiber activity associated with the sweat glands, hair follicles, and blood vessels of the perineum. Specific diseases or conditions that can be treated based on stimulation of a sympathetic nervous system include, for example, complex regional pain syndrome, peripheral vascular disease and chronic limb ischemia, angina

pain, diabetic pain, abdominal/visceral pain syndrome, phantom limb pain, Raynaud's syndrome, diabetic peripheral neuropathy, hypertension, hypotension, headache and migraine, and inflammatory pain such as arthritis, irritable bowel pain, osteoarthritis pain, and fibromyalgia. It should be appreciated that, in some embodiments, the electrode(s) **107** may be placed beside other autonomic structures including parasympathetic nerves (e.g., vagus nerve).

[0074] Referring now to FIG. **10**, the placement of the electrode or electrodes **107** (e.g., an electrode array) in order to deliver one or more electrical signals to an area within or adjacent target neural tissue, non-neural tissue, or a combination thereof **133** located adjacent or near a peripheral nerve is discussed in more detail. For instance, one or more electrodes **107** can be positioned near or adjacent a peripheral nerve at any location along its length, where the peripheral nerve can run, for instance, down the length of the leg **138** of the patient **116**. In the particular embodiment of FIG. **10**, the target tissue **133** is located adjacent the sciatic nerve **140**, although it is to be understood that neural tissue, non-neural tissue, or a combination thereof can be located adjacent any peripheral nerve in the leg (e.g., the common peroneal nerve **142**, the tibial nerve **144**, etc.), or any other location in the body. By placing the electrode or electrodes **107** adjacent or near a peripheral nerve, electrical signals can be delivered to the target neural tissue, non-neural tissue, or a combination thereof **133** to provide therapy to the patient. Specific diseases or conditions that can be treated based on stimulation of a peripheral nerve include, for example, acute pain, failed back surgery syndrome, complex regional pain syndrome, peripheral vascular disease and chronic limb ischemia, angina pain, diabetic pain, abdominal/visceral pain syndrome, brachial plexitis, phantom limb pain, intractable pain secondary to spinal cord injury, mediastinal pain, Raynaud's syndrome, headache and migraine, cervical neuritis, post-herpetic neuralgia, vertigo, tinnitus, hearing loss and inflammatory pain such as arthritis, irritable bowel pain, overactive bladder, bowel incontinence or constipation, osteoarthritis pain, and fibromyalgia. For example, electrical signals can be used to stimulate the sacral nerve roots to treat overactive bladder, fecal incontinence, and/or sexual dysfunction. In some embodiments, the electrode(s) may be placed adjacent or near a cranial nerve.

[0075] It should be appreciated from the description that the electrode(s) **107** may be placed in particular location in order to treat a particular condition using the electrical stimulation technologies described herein. For example, in an embodiment, the electrode(s) **107** may be placed within an epidural space between T7 and T12 of a thoracic portion of the patient's spine to treat the patient for spinal lumbar pain. In another embodiment, the electrode(s) **107** may be placed within an epidural space between C2 and T1 of a cervical portion of the patient's spine to treat the patient for spinal cervical pain. In another embodiment, the electrode(s) **107** may be placed within an epidural space between C2 and T8 of the patient's spine to treat angina pain. In another embodiment, the electrode(s) **107** may be placed within an epidural space between C2 and T8 of the patient's spine to treat abdominal pain. In another embodiment, the electrode(s) **107** may be placed within an epidural space between T10 and L5 of the patient's spine to treat peripheral vascular disease. In another embodiment, the electrode(s) **107** may be placed within an epidural space between T7 and T12 in the thoracic spine to treat spinal lumbar pain disorders. In another embodiment, the electrode(s) **107** may be placed within an epidural space between C2 and T1 of a cervical portion of the patient's spine to treat the patient for upper limb ischemia. In another embodiment, the electrode(s) **107** may be placed at or within a dorsal root ganglion of the patient's spine to treat chronic or acute pain. In another embodiment, the electrode(s) **107** may be placed within a sacral portion of the patient's spine to treat urinary or fecal incontinence. In various embodiments, the electrode(s) **107** may be placed near or around the lumbar sympathetic plexus, the celiac sympathetic plexus, the hypogastric sympathetic plexus, or the stellate ganglion to treat chronic or acute pain of the limb, abdomen, pelvic area, or upper extremity, respectively. In another embodiment, the electrode(s) **107** may be placed near or around the patient's brain to treat movement disorders, Parkinson's, pain, psychiatric and/or seizure disorders. In another embodiment, the electrode(s) **107** may be placed near or around the patient's vagus nerve to treat

seizure disorders, obesity, pain, or autonomic disorders. In another embodiment, the electrode(s) **107** may be placed near or around a peripheral nerve of the patient to treat acute pain, chronic pain, fecal or urinary incontinence, seizure disorders, movement disorders, obesity, spasticity, and to modulate other unpleasant neurological conditions. In another embodiment, the electrode(s) **107** may be placed near or around the patient's somatic tissue, muscles, connective tissue, or non-neural tissue to treat acute pain, chronic pain, fecal or urinary incontinence, seizure disorders, movement disorders, obesity, spasticity, and to modulate other unpleasant neurological conditions. In another embodiment, the electrode(s) **107** may be placed near or around the patient's visceral tissue or organs, and non-neural tissue to treat acute pain, chronic pain, fecal or urinary incontinence, seizure disorders, movement disorders, obesity, spasticity, and to modulate other unpleasant neurological conditions.

[0076] The various components of the systems **50**, **100**, and **300** described in FIGS. **1-10** may form the portion of an electrical stimulation system **400** as described below in more detail in reference to FIG. **11**.

[0077] Referring now to FIG. **11**, a simplified block diagram of at least one embodiment of an electrical stimulation system **400** for providing therapy to a patient via the application of one or more electrical signals and otherwise performing the functions described herein is shown. The illustrative electrical stimulation system **400** includes a plurality of electrodes **107**, a signal generator **109**, a controller **110**, a power system **111**, a user interface **112**, a patient monitoring system **404**, and a communication circuitry **406**. Further, in the illustrative embodiment, the controller **110** includes a processor **401**, an input/output (“I/O”) subsystem **402**, and a memory **403**, and the patient monitoring system **404** includes one or more sensors **405**. Additionally, as described herein, the electrodes **107** may collectively form one or more electrode arrays **408**. It should be appreciated that one or more of the components of the electrical stimulation system **400** described herein may be embodied as, or form a portion of, one or more embedded controllers and/or integrated circuits. For example, in some embodiments, the processor **401**, the I/O subsystem **402**, the memory **403** and/or other components of the electrical stimulation system **400** may be embodied as, or form a portion of, a microcontroller or SoC (e.g., such as an embodiment in which the controller **110** is a microcontroller). Further, depending on the particular embodiment, the components of the electrical stimulation system **400** may be closely positioned to one another or spatially distributed (i.e., separated from one another) depending on the particular embodiment. Additionally, although only a single generator **109**, controller **110**, power system **111**, user interface **112**, patient monitoring system **404**, communication circuitry **406**, electrode array **408**, processor **401**, I/O subsystem **402**, and memory **403** are illustratively shown in FIG. **11**, it should be appreciated that a particular electrical stimulation system **400** may include multiple signal generators **109**, controllers **110**, power systems **111**, user interfaces **112**, patient monitoring systems **404**, communication circuitries **406**, electrode arrays **408**, processors **401**, I/O subsystems **402**, and/or memories **403** in various embodiments. As described herein, it should be appreciated that the randomization of the stimulation can be managed by hardware, firmware, and/or software depending on the particular embodiment.

[0078] The electrodes **107** of the electrode array **408** can be used to deliver the electrical signals to the target neural tissue, non-neural tissue, or a combination thereof as described herein. Depending on the particular embodiment, the electrode array **408** can be implantable, percutaneous, or transcutaneous. Further, as described herein, the shape, size, material composition, inter-electrode spacing, and/or other parameters of the electrodes **107** or electrode array **408** can be specific to contouring the electrical field surrounding the target neural tissue, non-neural tissue, or a combination thereof, to enable specific therapy to be provided to the target neural tissue, non-neural tissue, or a combination thereof. It should be appreciated that the electrodes **107** of the electrode array **408** may be embodied on or electrically coupled to one or more electrical leads **108**, which may be connected to the signal generator **109** as described herein. Various embodiments of

the electrode array **408** are described in greater detail below in reference to FIGS. **13-20**. It should be appreciated that the electrode array **408**, the signal generator **109**, and/or the controller **110** include electrical circuitry that allow for the real-time selection of specific electrodes **107** of the electrode array **408** as being electrical sources, electrical sinks, or unused/off during the transmission of an electrical stimulation signal.

[0079] In some embodiments, one or more of the electrodes **107** may contain a thermistor for recording tissue temperature during stimulation, and for providing feedback information for efficacy and safety measures, and temperature control. One or more of the electrodes **107** may also be configured to measure tissue impedance for providing feedback information for efficacy and safety measures, and impedance control. Lastly, a cooling mechanism may be incorporated into the electrode design to control temperature at the electrode-tissue interface. Independent current channels, electrolytic jelly, and electrode insulation may be used to prevent co-excitation of surrounding tissue and to optimize the electrical field that is exposed to the nerve. The electrode array **408** may be placed through a needle-like introducer assembly or applicator, and may include catheter-over-needle or needle-over-catheter approaches where the catheter may include electrical contacts for delivery of the stimulation waveform. The position of specific geometric features of the electrodes **107** and/or electrode array **408** (e.g. tip, side, different contacts, etc.) relative to the target structure may be optimized to provide the desired therapeutic effect with minimal undesired side-effects (such as co-excitation of nearby excitable tissues).

[0080] As shown in the figures, the electrodes **107** (or electrode array **408**) may be connected to an implantable signal generator **109** through an electrical lead **108**. Alternatively, in some embodiments, the signal generator **109** can be external and can be wirelessly connected to the electrodes **107** (or electrode array **408**). In some embodiments, the signal generator **109** can be configured to generate and deliver electrical signals to provide therapy to a patient that can be customized based on patient feedback. As described herein, the patient feedback may be based on self-report (e.g., introspection, observations of unwanted sensory-and/or motor activity, autonomic dysfunction, etc.), determined from data generated by sensors measuring physiological outcomes, based on data from artificial intelligence or machine learning systems (e.g., output or feedback data), or based on combinations thereof.

[0081] The use of multiple signal generators **109** (or multiple waveform generators) may enhance the resolution of the signal generator **109** by independently powering some electrical contacts with one signal, and powering other contacts with a different signal, and/or by allowing one waveform to be added to another waveform and delivered through the same set of contacts. Accordingly, an electrical stimulation system **400** that contains multiple independent signal generators **109** (or multiple waveform generators) may enable improved control of electrical contacts, may enable improved control of the waveform's randomness, and/or may facilitate interactions between frequency bands (e.g., multi-waveform, physiological, and psychophysical masking).

[0082] Regardless of the particular type or combination of electrical signals utilized, the electrical signals can be adjusted, such that the energy contained within a particular frequency band, and for all frequency bands of energy delivered to the tissue, can be adjusted to best treat the patient as described herein. In illustrative embodiments, the adjustable electrical energy can be adjusted to deliver electrical signals with intensities ranging from about 1 mA (or 0.1 mA) to about 100 mA and from about 1 V (or 0.1 V) to about 200 V (peak-to-peak) for each frequency band included in the spectrum. In some embodiments, the spectrum of electrical signals includes frequencies ranging from about 0 Hz to about 100 kHz (or about 0 Hz to about 500 kHz), and is composed of adjustable frequency bands. It should be appreciated that the size of the frequency bands may vary depending on the particular embodiment (e.g., 10 Hz bands, 100 Hz bands, 1000 Hz bands, etc.). In some embodiments, the spectrum of electrical signals includes frequencies ranging from about 0 Hz to about 25 kHz (or about 0 Hz to about 25 kHz), and each of the frequency bands within the frequency range may have a bandwidth of 1 kHz or 2 kHz. In another embodiment, the electrical

stimulation signal may have a frequency range of about 0.05 Hz to about 2 kHz, and/or each of the frequency bands within the frequency range may have a bandwidth of about 150 Hz. Although various frequency ranges are described herein as being “from about 0 Hz,” it should be appreciated that such lower bound may be some non-zero frequency greater than 0 Hz in some embodiments (e.g., 1 Hz). Moreover, the frequency band that receives power can be determined in a random fashion.

[0083] The power supply, power source, or power system **111** is configured to supply power to the controller **110** and/or other components of the electrical stimulation system **400**. In some embodiments, the power system **111** is an independent, untethered, and portable power source configured to supply power to the electrical stimulation system **400** to perform the various functions described herein. For example, the power system **111** may include one or more batteries, battery packs, capacitors, super capacitors, solar cells, and/or other power supplies. Depending on the particular embodiment, the power system **111** may or may not be rechargeable. In other embodiments, the power system **111** may be line powered via AC mains and/or another suitable power source. It should be appreciated that the power system **111** can include both external and internal portions, where the internal portion of the power system can include a battery, such as a lithium battery, and the external portion of the power system **111** can be plugged into a wall and used to recharge the battery as needed. In such embodiments, the external portion of the power system **111** can transmit power to the signal generator **109** as directed by the controller **110** via RF signals/electromagnetic induction, or power can be transmitted to the signal generator **109** via the battery in the internal portion of the power system **111**. Further, the external portion of the power system **111** can be used to recharge the battery in the internal portion of the power system **111**.

[0084] The user interface **112** may be embodied as any one or more devices or components that allow a user to interact with the electrical stimulation system **400**. For example, in some embodiments, the user interface **112** can be in the form of a computer that interacts with the controller **110** and is powered by a power system **111**. In particular, in some embodiments, the computer can operate software designed to record signals passed from the controller **110**, and to drive the controller's output. Possible software packages include Cambridge Electronic Design's (UK) SPIKE program. The software can be programmable and can record and analyze electrophysiological signals, as well as direct the controller **110** to deliver the electrical signals described herein. Further, in some embodiments, the user interface **112** may include one or more peripheral devices such as, for example, a keyboard, mouse, display, status indicator, diagnostic tool, speaker, microphone, and/or one or more other suitable peripheral devices.

[0085] In some embodiments, the electrical stimulation system **400** may include a patient monitoring system **404**. In such embodiments, the patient monitoring system **404** can acquire, amplify/attenuate, and filter physiological signals and then output them to the controller **110**. It should be appreciated that the patient monitoring system **404** may include one or more sensors **405**. The sensors **405** are configured to generate sensor data (e.g., by virtue of one or more signals), which may be interpreted by the controller **110** (e.g., the processor **401**) to determine one or more characteristics associated with the patient and/or the electrical stimulation system **400**. By way of example, the sensors **405** may detect various characteristics of the physical environment of the electrical stimulation system **400** (internal and/or external) and/or other suitable characteristics. In various embodiments, the sensors **405** may be embodied as, or otherwise include, environmental sensors, inertial sensors, proximity sensors, optical sensors, electromagnetic sensors, audio sensors, pressure sensors, flow meters, temperature sensors, thermistors, chemical sensors, biopotential electrodes, motion sensors, piezoelectric sensors, cameras, and/or other types of sensors. Of course, the electrical stimulation system **400** may also include components and/or devices configured to facilitate the use of the sensors **405**. For example, in some embodiments, the patient monitoring system **404** may include a heart-rate monitor to collect electrocardiogram signals and/or a muscle activity monitor to collect electromyography signals. The heart-rate monitor can include ECG

electrodes coupled with an alternating current (AC) amplifier, and the muscle activity monitor can include EMG electrodes coupled with an AC amplifier. Other types of transducers may also be used in other embodiments. As described, physiological signals obtained with the patient monitoring system **404** may be passed through an AC signal amplifier/conditioner. One possible amplifier/conditioner is a Model LP511 AC amplifier available from Grass Technologies, a subsidiary of Astro-Med, Inc., West Warwick, Rhode Island, USA.

[0086] The communication circuitry **406** may be embodied as any communication circuitry, transceiver, device, or collection thereof, capable of enabling communications between the electrical stimulation system **400** and other remote devices. The communication circuitry **406** may be configured to use any one or more wired and/or wireless communication technologies and associated protocols. For example, in some embodiments, the illustrative electrical stimulation system **400** may be configured to communicate via Wi-Fi (e.g., infrastructure or ad hoc mode), Wi-Fi Direct, Bluetooth (including Bluetooth Low Energy (BLE)), ZigBee, Z-wave, Near Field Communication (NFC), IEEE 802.15, HL7, and/or other suitable wireless communication protocol(s). Further, in some embodiments, the electrical stimulation system **400** may be configured to communicate via Ethernet, Power over Ethernet (PoE), serial communication links, power line communication, and/or another suitable wired communication mechanism.

[0087] The controller **110** may be embodied as any type of controller or control system capable of performing the functions described herein. In the illustrative embodiment, the controller **110** can record electrical signal data as well as digital information from the patient monitoring system **404**, and can generate electrical signal and digital outputs simultaneously for real-time control of the signal generator **109** based on feedback received from the patient after transmission of the electrical stimulation signals. The controller **110** may have onboard memory to facilitate high-speed data capture, independent waveform sample rates, and on-line analysis. An exemplary controller **110** may be a POWER **1401** data-acquisition interface unit available from Cambridge Electronic Design (UK).

[0088] As shown, in some embodiments, the controller **110** includes a processor **401**, an I/O subsystem **402**, and memory **403**.

[0089] The processor **401** may be embodied as any type of processor(s) capable of performing the functions described herein. In particular, the processor **401** may be embodied as one or more single or multi-core processors, microcontrollers, or other processor or processing/controlling circuits. For example, in some embodiments, the processor **401** may include or be embodied as an arithmetic logic unit (ALU), central processing unit (CPU), digital signal processor (DSP), and/or another suitable processor(s). The processor **401** may be a programmable type, a dedicated hardwired state machine, or a combination thereof. One or more processors **401** with multiple processing units may utilize distributed, pipelined, and/or parallel processing in various embodiments. Further, the processor **401** may be dedicated to performance of just the operations described herein, or may be utilized in one or more additional applications. In the illustrative embodiment, the processor **401** is of a programmable variety that executes algorithms and/or processes data in accordance with operating logic as defined by programming instructions (such as software or firmware) stored in the memory **403**. Additionally or alternatively, the operating logic for the processor **401** may be at least partially defined by hardwired logic or other hardware. Further, the processor **401** may include one or more components of any type suitable to process the signals received from input/output devices or from other components or devices and to provide desired output signals. Such components may include digital circuitry, analog circuitry, or a combination thereof.

[0090] The memory **403** may be of one or more types of non-transitory computer-readable media, such as a solid-state memory, electromagnetic memory, optical memory, or a combination thereof. Furthermore, the memory **403** may be volatile and/or nonvolatile and, in some embodiments, some or all of the memory **403** may be of a portable variety, such as a disk, tape, memory stick, cartridge, and/or other suitable portable memory. In operation, the memory **403** may store various data and

software used during operation of the electrical stimulation system **400** such as operating systems (e.g., real-time operating systems (RTOS)), applications, programs, libraries, and drivers. The memory **403** is communicatively coupled to the processor **401** via the I/O subsystem **402**, which may be embodied as circuitry and/or components to facilitate input/output operations with the processor **401**, the memory **403**, and other components of the electrical stimulation system **400**. For example, the I/O subsystem **402** may be embodied as, or otherwise include, memory controller hubs, input/output control hubs, firmware devices, communication links (i.e., point-to-point links, bus links, wires, cables, light guides, printed circuit board traces, etc.) and/or other components and subsystems to facilitate the input/output operations. Depending on the particular embodiment, the memory **403** may be included with the processor **401** and/or coupled to the processor **401** depending on the particular embodiment. For example, in some embodiments, the processor **401**, the I/O subsystem **402**, the memory **403**, and/or other components of the electrical stimulation system **400** may form a portion of a system-on-a-chip (SoC) and be incorporated on a single integrated circuit chip.

[0091] As shown in FIG. **11**, in some embodiments, the electrical stimulation system **400** or a portion thereof (e.g., the controller **110**) may be configured to communicate with an analytics server **420** and/or other remote computing device via a network **410**. For example, in some embodiments, the electrical stimulation system **400** may transmit patient data (e.g., including an optimal electrical stimulation signature), sensor data, and/or other data to the analytics server **420** for leveraging artificial intelligence, machine learning, and/or other technologies for pattern identification and/or decision-making.

[0092] The network **410** may be embodied as any type of communication network capable of facilitating communication between the electrical stimulation system **400** and the analytics server **420** and/or other remote devices. As such, the network **410** may include one or more networks, routers, switches, computers, and/or other intervening devices. For example, the network **410** may be embodied as or otherwise include one or more cellular networks, telephone networks, local or wide area networks, publicly available global networks (e.g., the Internet), ad hoc networks, short-range communication links, or a combination thereof.

[0093] The analytics server **420** may be embodied as any type of device(s) capable of performing the functions described herein. It should be appreciated that the efficacy of neuro-modulation technologies is affected by continuously changing treatment variables. Treatment variables may be device-specific (e.g., lead migration and impedance changes, stimulation paradigm, etc.), physiological (e.g., neurological conditioning or tolerance, scar tissue formation, plasticity, etc.), psychological (e.g., depression, etc.), disease state specific (e.g., progression, improvement, etc.) and/or patient dependent (e.g., height, weight, age, race, etc.). Moreover, these treatment variables generally change with time and may interact (stim*time interaction, long-term physiological changes, etc.).

[0094] In some embodiments, the inputs for a neural network or other machine learning algorithm used by the analytics server **420** may include one or more of adjusted electrical stimulation signatures, patient information, patient demographics, diagnosis information, electrode positioning, patient sensations, measures of treatment efficacy and/or outcomes (e.g., Pain Rating Scale scores), time of day, duration of treatment, time elapsed since start of treatment plan, and/or other machine learning model inputs. Further, in various embodiments, the analytics server **420** may utilize any machine learning and/or artificial intelligence algorithm for performing the functions described herein. For example, in some embodiments, the analytics server **420** may utilize one or more neural network algorithms, regression algorithms, instance-based algorithms, regularization algorithms, decision tree algorithms, Bayesian algorithms, clustering algorithms, association rule learning algorithms, deep learning algorithms, dimensionality reduction algorithms, and/or other suitable machine learning algorithms, techniques, and/or mechanisms.

[0095] It should be further appreciated that, although the analytics server **420** is described herein as

a computing device outside of a cloud computing environment, in other embodiments, the analytics server **420** may be embodied as a cloud-based device or collection of devices within a cloud computing environment. Further, in cloud-based embodiments, the analytics server **420** may be embodied as a server-ambiguous computing solution, for example, that executes a plurality of instructions on-demand, contains logic to execute instructions only when prompted by a particular activity/trigger, and does not consume computing resources when not in use. That is, the analytics server **420** may be embodied as a virtual computing environment residing “on” a computing system (e.g., a distributed network of devices) in which various virtual functions (e.g., Lambda functions, Azure functions, Google cloud functions, and/or other suitable virtual functions) may be executed corresponding with the functions of the analytics server **420** described herein. For example, when an event occurs (e.g., data is transferred to the analytics server **420** for handling), the virtual computing environment may be communicated with (e.g., via a request to an API of the virtual computing environment), whereby the API may route the request to the correct virtual function (e.g., a particular server-ambiguous computing resource) based on a set of rules. For example, when a request for the transmission of data is made (e.g., via an appropriate user interface to the analytics server **420**), the appropriate virtual function(s) may be executed to perform the actions before eliminating the instance of the virtual function(s).

[0096] Referring now to FIG. **12**, a simplified block diagram of at least one embodiment of a computing device **500** is shown. The illustrative computing device **500** depicts at least one embodiment of a server that may be utilized in connection with the analytics server **420** illustrated in FIG. **12** and/or other devices in communication with the electrical stimulation system **400**. Depending on the particular embodiment, the computing device **500** may be embodied as a server, desktop computer, laptop computer, tablet computer, notebook, netbook, Ultrabook™, mobile computing device, cellular phone, smartphone, wearable computing device, personal digital assistant, Internet of Things (IoT) device, processing system, router, gateway, and/or any other computing, processing, and/or communication device capable of performing the functions described herein.

[0097] The computing device **500** includes a processing device **502** that executes algorithms and/or processes data in accordance with operating logic **508**, an input/output device **504** that enables communication between the computing device **500** and one or more external devices **510**, and memory **506** which stores, for example, data received from the external device **510** via the input/output device **504**.

[0098] The input/output device **504** allows the computing device **500** to communicate with the external device **510**. For example, the input/output device **504** may include a transceiver, a network adapter, a network card, an interface, one or more communication ports (e.g., a USB port, serial port, parallel port, an analog port, a digital port, VGA, DVI, HDMI, FireWire, CAT 5, or any other type of communication port or interface), and/or other communication circuitry. Communication circuitry of the computing device **500** may be configured to use any one or more communication technologies (e.g., wireless or wired communications) and associated protocols (e.g., Ethernet, Bluetooth®, Wi-Fi®, WiMAX, etc.) to effect such communication depending on the particular computing device **500**. The input/output device **504** may include hardware, software, and/or firmware suitable for performing the techniques described herein.

[0099] The external device **510** may be any type of device that allows data to be inputted or outputted from the computing device **500**. For example, in various embodiments, the external device **510** may be embodied as the analytics server **420** and/or the electrical stimulation system **400**. Further, in some embodiments, the external device **510** may be embodied as another computing device, switch, diagnostic tool, controller, printer, display, alarm, peripheral device (e.g., keyboard, mouse, touch screen display, etc.), and/or any other computing, processing, and/or communication device capable of performing the functions described herein. Furthermore, in some embodiments, it should be appreciated that the external device **510** may be integrated into the

computing device **500**.

[0100] The processing device **502** may be embodied as any type of processor(s) capable of performing the functions described herein. In particular, the processing device **502** may be embodied as one or more single or multi-core processors, microcontrollers, or other processor or processing/controlling circuits. For example, in some embodiments, the processing device **502** may include or be embodied as an arithmetic logic unit (ALU), central processing unit (CPU), digital signal processor (DSP), and/or another suitable processor(s). The processing device **502** may be a programmable type, a dedicated hardwired state machine, or a combination thereof. Processing devices **502** with multiple processing units may utilize distributed, pipelined, and/or parallel processing in various embodiments. Further, the processing device **502** may be dedicated to performance of just the operations described herein, or may be utilized in one or more additional applications. In the illustrative embodiment, the processing device **502** is programmable and executes algorithms and/or processes data in accordance with operating logic **508** as defined by programming instructions (such as software or firmware) stored in memory **506**. Additionally or alternatively, the operating logic **508** for processing device **502** may be at least partially defined by hardwired logic or other hardware. Further, the processing device **502** may include one or more components of any type suitable to process the signals received from input/output device **504** or from other components or devices and to provide desired output signals. Such components may include digital circuitry, analog circuitry, or a combination thereof.

[0101] The memory **506** may be of one or more types of non-transitory computer-readable media, such as a solid-state memory, electromagnetic memory, optical memory, or a combination thereof. Furthermore, the memory **506** may be volatile and/or nonvolatile and, in some embodiments, some or all of the memory **506** may be of a portable type, such as a disk, tape, memory stick, cartridge, and/or other suitable portable memory. In operation, the memory **506** may store various data and software used during operation of the computing device **500** such as operating systems, applications, programs, libraries, and drivers. It should be appreciated that the memory **506** may store data that is manipulated by the operating logic **508** of processing device **502**, such as, for example, data representative of signals received from and/or sent to the input/output device **504** in addition to or in lieu of storing programming instructions defining operating logic **508**. As shown in FIG. 12, the memory **506** may be included with the processing device **502** and/or coupled to the processing device **502** depending on the particular embodiment. For example, in some embodiments, the processing device **502**, the memory **506**, and/or other components of the computing device **500** may form a portion of a system-on-a-chip (SoC) and be incorporated on a single integrated circuit chip.

[0102] In some embodiments, various components of the computing device **500** (e.g., the processing device **502** and the memory **506**) may be communicatively coupled via an input/output subsystem, which may be embodied as circuitry and/or components to facilitate input/output operations with the processing device **502**, the memory **506**, and other components of the computing device **500**. For example, the input/output subsystem may be embodied as, or otherwise include, memory controller hubs, input/output control hubs, firmware devices, communication links (i.e., point-to-point links, bus links, wires, cables, light guides, printed circuit board traces, etc.) and/or other components and subsystems to facilitate the input/output operations.

[0103] The computing device **500** may include other or additional components, such as those commonly found in a typical computing device (e.g., various input/output devices and/or other components), in other embodiments. It should be further appreciated that one or more of the components of the computing device **500** described herein may be distributed across multiple computing devices. In other words, the techniques described herein may be employed by a computing system that includes one or more computing devices. Additionally, although only a single processing device **502**, I/O device **504**, and memory **506** are illustratively shown in FIG. 12, it should be appreciated that a particular computing device **500** may include multiple processing

devices **502**, I/O devices **504**, and/or memories **506** in other embodiments. Further, in some embodiments, more than one external device **510** may be in communication with the computing device **500**.

[0104] As described above, the electrical stimulation system **400** is configured to randomly deliver electrical stimuli through a set of electrodes in an electrode array **408** to neural and/or non-neural tissue, in order to more effectively treat acute/chronic pain and/or other medical conditions while avoiding neurological tolerance in the patient. More specifically, in various illustrative embodiments, the electrical stimulation system **400** utilizes an electrode array **408** including a plurality of electrodes **107** (e.g., three or more) that delivers spatially random electrical stimulation to the patient. Various embodiments of the electrode array **408** and corresponding electrical configurations are depicted and described in reference to FIGS. **13-20**.

[0105] Referring now to FIG. **13**, at least one embodiment of an electrode array **1300** is shown as part of an electrode assembly **1302**. As shown, the illustrative electrode array **1300** includes eight electrodes **107**, which are labeled as E1-E8 for convenience. Although the illustrative embodiment of the electrode array **1300** depicts eight electrodes **107**, it should be appreciated that the electrode array **1300** may include a greater or lesser number of electrodes in other embodiments. It should be appreciated that the electrode array **1300** (or, more specifically, the electrodes **107** thereof) may be electrically coupled to a signal generator **109** through a lead **108** as described herein. As shown, in the illustrative embodiment, the electrode assembly **1302** is relatively rectangular in cross-sectional shape, extending lengthwise and distally from a point of connection **1304** with the lead **108** (e.g., along a y-axis). However, it should be appreciated that the electrode assembly **1302** may be otherwise shaped in other embodiments (e.g., cylindrical). In the illustrative embodiment, each of the electrodes **107** (i.e., E1-E8) extends laterally across the electrode assembly **1302** (e.g., along an x-axis) and is spatially separated from the other electrodes **107** (e.g., to avoid electrical interference between electrodes **107** and/or other unwanted electrical/electromagnetic characteristics). In the illustrative embodiment, the E1 electrode **107** is depicted as the most distal electrode **107**, and the E8 electrode **108** is depicted as the most proximal electrode **107**. In other embodiments, however, it should be appreciated that the lead **108** may otherwise connect with the electrode assembly **1302**.

[0106] In use, in the illustrative embodiment, the controller **110** of the electrical stimulation system **400** is configured to randomly select a set of one or more electrodes **107** (e.g., from E1-E8) to function as electrical sources and a separate set of one or more electrodes **107** (e.g., from E1-E8) to function as electrical sinks. It should be appreciated that the remaining electrodes **107** (e.g., from E1-E8) that have not been selected to function as electrical sources or electrical sinks may remain off or otherwise non-functioning. Further, in some embodiments, the number of electrodes **107** selected as sources and/or the number of electrodes **107** selected as sinks may also be randomly determined. In some embodiments, the controller **110** may also randomly select the polarity of the assignment. It should be appreciated that the electrical stimulation system **400** may use any suitable combination of hardware, firmware, and/or software and/or any suitable algorithm or technique to perform the random selections described herein.

[0107] By way of example, FIGS. **14-17** illustrate various electrical configurations of the electrode array **1300** that could be randomly selected by the controller **110**. FIG. **14** depicts a configuration in which the controller **110** has randomly selected the E1 electrode **107** to be a source **1402** and the E3 electrode **107** to be a sink **1404**. FIG. **15** depicts a configuration in which the controller **110** has randomly selected the E1, E3, and E4 electrodes **107** to be sources **1402** and the E2 and E5 electrodes **107** to be sinks **1404**. FIG. **16** depicts a configuration in which the controller **110** has randomly selected the E1, E2, E3, and E4 electrodes **107** to be sources **1402** and the E7 and E8 electrodes **107** to be sinks **1404**. FIG. **17** depicts a configuration in which the controller **110** has randomly selected the E4 and E5 electrodes **107** to be sources **1402** and the E1 and E2 electrodes to be sinks **1404**.

[0108] It should be appreciated that the amount of time that a particular signal is delivered to the

patient through the selected electrical configuration of source electrode(s) **107** and sink electrode(s) **107** may vary depending on the particular embodiment. For example, in some embodiments, the controller **110** may instruct the signal generator **109** to deliver the signal through the selected electrical configuration of source electrode(s) **107** and sink electrode(s) **107** for a threshold period of time, after which the controller **110** again randomly selects a new electrical configuration of source electrode(s) **107** and sink electrode(s) **107** and instructs the signal generator **109** to deliver the same signal (or different signal) through the newly selected electrical configuration of source electrode(s) **107** and sink electrode(s) **107** for the threshold period of time. Further, it should be appreciated that the controller **110** may utilize any suitable technique for measuring time and determining the threshold period of time (e.g., number of clock cycles, number of pulses for a pulse-based signal, number of periods for a periodic signal, portions thereof, etc.). For example, in an embodiment, the controller **110** may instruct the signal generator **109** to deliver a set of pulses to the patient in which a first pulse is delivered through the electrode array **1300** using the electrical configuration of FIG. **15**, a second pulse is delivered through the electrode array **1300** using the electrical configuration of FIG. **16**, and a third pulse is delivered through the electrode array **1300** using the electrical configuration of FIG. **17**. Further, in some embodiments, the controller **110** may utilize sensed data and/or feedback to determine the amount of time that a particular signal is delivered to the patient through the selected electrical configuration of source electrode(s) **107** and sink electrode(s) **107**. For example, the relevant threshold may be based on the amount of charge injected by the electrode(s) **107**, charge per pulse, charge density, feedback from the controller **110** (e.g., based on previous stimulation history), and/or other characteristics.

[0109] As described herein, in various embodiments, the electrical stimulation signal may be a periodic signal, aperiodic signal, impulses, pulses, direct current waveform, noise signal, and/or another waveform (e.g., 0-1 MHz). Further, in some embodiments, the electrical stimulation signal has been tuned (or is being tuned) to treat the patient by adjusting various characteristics of the signal (e.g., adjusting the amplitude of the voltage/current, power, and/or phase of the signal within various frequency bands).

[0110] Referring now to FIG. **18**, at least one embodiment of an electrode array **1800** is shown as part of an electrode assembly **1802**. As shown, the illustrative electrode array **1800** includes twenty-seven electrodes **107**, which are divided into three groups of nine electrodes **107**. Specifically, nine electrodes **107** are grouped together at a distal section **1804** of the electrode assembly **1802**, nine electrodes **107** are grouped together at a central section **1806** of the electrode assembly **1802**, and nine electrodes **107** are grouped together at a proximal section **1808** of the electrode assembly **1802**. For clarity of the figure, only a subset of the twenty-seven electrodes **107** have been expressed labeled in FIG. **18**; however, such designation is for reference only and not intended to convey a distinction. Similar conventions have been employed in FIGS. **19** and **20**. Although the illustrative embodiment of the electrode array **1800** depicts twenty-seven electrodes **107**, it should be appreciated that the electrode array **1800** may include a greater or lesser number of electrodes in other embodiments. Further, although the illustrative embodiment of the electrode array **1800** depicts three groups of electrodes **107** (i.e., sections **1804**, **1806**, **1808**), it should be appreciated that the electrode array **1800** may include a greater or lesser number of sections in other embodiments. Additionally, it should be appreciated that the number of electrodes **107** in a particular section of the electrode array **1800** (e.g., sections **1804**, **1806**, **1808**) may vary depending on the particular embodiment, and may differ between sections. For example, in some embodiments, section **1804** may include a different number of electrodes **107** from section **1806** and/or section **1808**. Additionally, although the sections **1804**, **1806**, **1808** of the electrode assembly **1802** are described in reference to their distal/proximal location on the electrode assembly **1802**, it should be appreciated that electrodes **107** may be otherwise grouped based on their locations in other embodiments. For example, in some embodiments, the electrode assembly **1802** may include medial and lateral sections and, therefore, corresponding groups of electrodes

107.

[0111] It should be appreciated that the electrode array **1800** (or, more specifically, the electrodes **107** thereof) may be electrically coupled to a signal generator **109** through a lead **108** as described herein. As shown, in the illustrative embodiment, the electrode assembly **1802** is relatively rectangular in cross-sectional shape, extending lengthwise and distally from a point of connection **1810** with the lead **108** (e.g., along a y-axis). However, it should be appreciated that the electrode assembly **1802** may be otherwise shaped in other embodiments (e.g., cylindrical). In the illustrative embodiment, each of the electrodes **107** has a circular cross-section and is spatially separated from the other electrodes **107** (e.g., to avoid electrical interference between electrodes **107** and/or other unwanted electrical/electromagnetic characteristics). In other embodiments, however, it should be appreciated that the lead **108** may otherwise connect with the electrode assembly **1802**.

[0112] In use, in the illustrative embodiment, the controller **110** of the electrical stimulation system **400** is configured to randomly select a set of one or more electrodes **107** of the electrode array **1800** to function as electrical sources and a separate set of one or more electrodes **107** to function as electrical sinks. It should be appreciated that the remaining electrodes **107** that have not been selected to function as electrical sources or electrical sinks may remain off or otherwise non-functioning. Further, in some embodiments, the number of electrodes **107** selected as sources and/or the number of electrodes **107** selected as sinks may also be randomly determined. In some embodiments, the controller **110** may also randomly select the polarity of the assignment. Further, in some embodiments, the separation distance between electrodes **107** of opposite polarity may be specified.

[0113] By way of example, FIG. **19** illustrates various electrical configurations of the electrode array **1800** that could be randomly selected by the controller **110**. FIG. **19** depicts at least one configuration in which the controller **110** has randomly selected two electrodes **107** in the distal section **1804** of the electrode array **1800** to be sources **1812**, two electrodes **107** in the central section **1806** of the electrode array **1800** to be sources **1812**, two electrodes **107** in the proximal section **1808** of the electrode array **1800** to be sources **1812**, two electrodes **107** in the distal section **1804** of the electrode array **1800** to be sinks **1814**, one electrode **107** in the central section **1806** of the electrode array **1800** to be a sink **1814**, and two electrodes **107** in the proximal section **1808** of the electrode array **1800** to be sinks. Although each of the sources **1812** and sinks **1814** is not separately labeled in FIG. **19**, for clarity, each of the sources **1812** is depicted in FIG. **19** as having the same fill pattern and each of the sinks **1814** is depicted in FIG. **19** as having a different fill pattern.

[0114] In some embodiments, the electrical stimulation and/or electrode assignments and configuration may be adjusted (or tuned) with respect to each of the sections **1804**, **1806**, **1808** of the electrode array **1800**. For example, the stimulation parameter (e.g., amplitude, frequency, pulse duration, charge per phase, etc.) and/or electrode assignment/selection delivered to each section **1804**, **1806**, **1808** may be tuned based on feedback. For example, in some embodiments, the distal section **1804** of the electrode array **1800** may be restricted to delivering high frequency signals (or high-frequency components of the electrical stimulation signals), whereas the central section **1806** and the proximal section **1808** may be restricted to delivering moderate and low frequency signals (or moderate-and low-frequency components of the electrical stimulation signals). In another embodiment, the central section **1806** may be restricted to delivering moderate frequency signals (or moderate-frequency components of the electrical stimulation signals) and the proximal section **1808** may be restricted to delivering low frequency signals (or low-frequency components of the electrical stimulation signals).

[0115] In another embodiment, the amplitude of the electrical stimulation signal being delivered may be independently adjusted or weighted for each of the sections **1804**, **1806**, **1808** of the electrode array **1800**. For example, in an embodiment, the amplitude of the electrical stimulation signal being delivered to the distal section **1804** of the electrode array **1800** may be

amplified/weighted by a factor of 2.0 (i.e., doubled), the amplitude of the electrical stimulation signal being delivered to the central section **1806** of the electrode array **1800** may be amplified/weighted by a factor of 0.5 (i.e., attenuated by one half), and the amplitude of the electrical stimulation signal being delivered to the proximal section **1808** of the electrode array **1800** may be amplified/weighted by a factor of 1.0 (i.e., passed through). Although the independent adjustment/weighting of the amplitude of the electrical stimulation signal is described herein as being done with respect to different sections **1804**, **1806**, **1808** of the electrode array **1800**, it should be appreciated that the amplitude of the electrical stimulation signal may be independently adjusted/weighted for each electrode **107** or other set(s) of electrodes **107** in other embodiments. [0116] Referring now to FIG. **20**, at least one embodiment of an electrode array **2000** is shown as part of an electrode assembly **2002**. As shown, the illustrative electrode array **2000** includes twenty-four electrodes **107** of varying sizes and shapes. Specifically, the electrode array **2000** includes fourteen electrodes **107**, **2004** that have a circular cross-section and are of a first size, eight electrodes **107**, **2006** that have a circular cross-section and are of a second size (i.e., smaller than the first size), and two electrodes **107**, **2008** that have an elliptical/oval cross-section and are of a third size (i.e., larger than the first size). Although the illustrative embodiment of the electrode array **2000** depicts twenty-four electrodes **107**, it should be appreciated that the electrode array **2000** may include a greater or lesser number of electrodes in other embodiments. Further, although the illustrative embodiment of the electrode array **2000** depicts three different sizes/shapes of electrodes **107** (i.e., electrodes **2004**, **2006**, **2008**), it should be appreciated that the electrode array **2000** may include a greater or lesser number of different sizes/shapes in other embodiments. Additionally, it should be appreciated that the number of electrodes **107** of a particular size/shape in the electrode array **2000** may vary depending on the particular embodiment.

[0117] It should be appreciated that the electrode array **2000** (or, more specifically, the electrodes **107** thereof) may be electrically coupled to a signal generator **109** through a lead **108** as described herein. As shown, in the illustrative embodiment, the electrode assembly **2002** is relatively rectangular in cross-sectional shape, extending lengthwise and distally from a point of connection **2010** with the lead **108** (e.g., along a y-axis). However, it should be appreciated that the electrode assembly **2002** may be otherwise shaped in other embodiments (e.g., cylindrical). In the illustrative embodiment, each of the electrodes **107** is spatially separated from the other electrodes **107** (e.g., to avoid electrical interference between electrodes **107** and/or other unwanted electrical/electromagnetic characteristics). In other embodiments, however, it should be appreciated that the lead **108** may otherwise connect with the electrode assembly **2002**.

[0118] In use, in the illustrative embodiment, the controller **110** of the electrical stimulation system **400** is configured to randomly select a set of one or more electrodes **107** of the electrode array **2000** to function as electrical sources and a separate set of one or more electrodes **107** to function as electrical sinks. It should be appreciated that the remaining electrodes **107** that have not been selected to function as electrical sources or electrical sinks may remain off or otherwise non-functioning. Further, in some embodiments, the number of electrodes **107** selected as sources and/or the number of electrodes **107** selected as sinks may also be randomly determined. In some embodiments, the controller **110** may also randomly select the polarity of the assignment.

[0119] By way of example, FIG. **20** illustrates at least one electrical configuration of the electrode array **2000** that could be randomly selected by the controller **110**. More specifically, FIG. **20** depicts at least one configuration in which the controller **110** has randomly selected two of the electrodes **107**, **2004** that have a circular cross-section and are of the first size as sources **2012**, one of the electrodes **107**, **2004** that have a circular cross-section and are of the first size as a sink **2014**, two of the electrodes **107**, **2006** that have a circular cross-section and are of the second size as sources **2012**, two of the electrodes **107**, **2006** that have a circular cross-section and are of the second size as sinks **2014**, one of the electrodes **107**, **2008** that have an elliptical/oval cross-section and are of the third size as a source **2012**, and one of the electrodes **107**, **2008** that have an

elliptical/oval cross-section and are of the third size as a sink **2014**. Although each of the sources **2012** and sinks **2014** is not separately labeled in FIG. **20**, for clarity, each of the sources **2012** is depicted in FIG. **20** as having the same fill pattern and each of the sources **2014** is depicted in FIG. **20** as having a different fill pattern.

[0120] It should be appreciated that the sizes and shapes of the electrodes **107** depicted in FIG. **20** are for illustrative purposes. Therefore, in various embodiments, a particular electrode array **408** may be designed to have different sized and/or shaped (e.g., circular, elliptical/oval, square, rectangular, polygonal, etc.) electrodes **107**. Further, in various embodiments, the electrodes **107** of the electrode array **408** may be composed of one or more materials (e.g., platinum, carbon, stainless steel, P-dot coated, roughed surface), which may vary among electrodes **107** in a particular electrode array **408**. In some embodiments, the electrical stimulation and/or electrode assignments and configuration may be adjusted (or tuned) with respect to the size, shape, and/or material of the various electrodes **107** of the electrode array **408**. For example, in some embodiments, low frequency signals (or low-frequency components of the electrical stimulation signals) may be restricted to being delivered through the largest electrodes **107** or the electrodes **107** with enhanced charge capacity characteristics, whereas the higher frequency signals (or higher-frequency components of the electrical stimulation signals) may be delivered on smaller sized electrodes **107** (or larger electrodes **107**).

[0121] Referring now to FIG. **21**, in use, the electrical stimulation system **400** may execute a method **2100** for randomly selecting an electrical configuration of an electrode array for delivering electrical stimulation to a patient. It should be appreciated that the particular blocks of the method **2100** are illustrated by way of example, and such blocks may be combined or divided, added or removed, and/or reordered in whole or in part depending on the particular embodiment, unless stated to the contrary.

[0122] The illustrative method **2100** begins with block **2102** in which the electrical stimulation system **400** or, more specifically, the signal generator **109** generates an electrical stimulation signal (e.g., in response to instructions from the controller **110**). As described above, in various embodiments, the electrical stimulation signal may be a periodic signal, aperiodic signal, impulses, pulses, direct current waveform, noise signal, and/or another waveform (e.g., 0-1 MHz). For example, in some embodiments, the electrical stimulation signal may have a frequency range of 0.05 Hz to 750 kHz. More specifically, in some embodiments, the electrical stimulation signal may have a frequency range of 0.05 Hz to 2 kHz, which blocks the action potential from even firing. In another embodiment, the electrical stimulation signal may have a frequency range of 10 kHz to 16 kHz, which stimulates the nerve so that it attenuates or prevents the nerve from firing a pain signal. In yet another embodiment, the electrical stimulation signal may have a frequency range of 500 kHz to 750 kHz, which results in so much energy around the nerve that it causes the nerve cell to run out of its mitochondrial stores of energy. In other words, such an electrical stimulation signal may fatigue the nerve, so that it essentially becomes tired and is unable to fire. As such, the electrical stimulation system **400** is able to deliver the signal to the patient for a period (e.g., five minutes), turn the stimulation off for an extended period (e.g., several days) while the nerve cells rebuild their mitochondrial stores, and again deliver the signal to the patient to again fatigue the nerve and deplete the stores. It has been discovered that delivering the electrical stimulation signal within this frequency range for approximately five minutes may block the nerve for approximately twenty-one days. Accordingly, it should be appreciated that the mechanism of action of the electrical stimulation signal that is randomly distributed across the electrodes **107** of the electrode array **408** may vary depending on the particular frequency range of the signal. In alternative embodiments, the electrical stimulation signal may have a frequency range of 1 kHz to 100 kHz, or the electrical stimulation signal may have a frequency range of 100 kHz to 1 MHz. It should be appreciated that the frequency range of 100 kHz to 1 MHz may selectively inhibit the parts of the nerve responsible for normal pain and autonomic function. The time-course of these effects have

been demonstrated to last days-to-weeks (e.g., 21 days post-treatment) when healthy neuronal circuitry is simulated, and weeks-to-months when stimulation is delivered to diseased neural circuitry (e.g., hosting chronic pain).

[0123] In block **2104**, the electrical stimulation system **400** (e.g., the controller **110**) randomly selects one or more electrodes **107** of the electrode array **408** to function as electrical sources and one or more other electrodes **107** of the electrode array **408** to function as electrical sinks. As described above, it should be appreciated that the number of electrodes **107** selected as electrical sources and/or the number of electrodes **107** selected as electrical sinks may also be randomly determined. Further, in some embodiments, the polarity of the assignment may also be randomly selected. In block **2106**, the electrical stimulation system **400** (e.g., the signal generator **109** by virtue of instructions received from the controller **110**) delivers the electrical stimulation signal to the patient through the electrode array **408** using the randomly selected electrical configuration. As indicated above, in some embodiments, it should be appreciated that various frequency components (e.g., high-, low-, or moderate-frequency components) of the electrical stimulation signal may further be directed to pass through a subset of the randomly selected electrodes of the electrode array **408**. Further, in some embodiments, the source electrode(s) **107** and sink electrode(s) **107** may be selected based on stimulus feature (e.g., frequency, amplitude, charge density, time, pulse number, etc.). For example, certain features/components of the electrical stimulation signal (e.g., a particular frequency range, amplitude, etc.) may be delivered to the patient through one set of electrode(s) **107**, whereas other features/components of the electrical stimulation signal may be delivered to the patient through another set of electrode(s) **107**.

[0124] As indicated above, in some embodiments, a particular electrical stimulation signal is only delivered to the patient through the selected electrical configuration of source electrode(s) **107** and sink electrode(s) **107** for a threshold period of time (or until another threshold is met), which may be measured using various techniques. Accordingly, in block **2108**, the electrical stimulation system **400** determines whether the threshold has been met. If not, the method **2100** returns to block **2106** in which the electrical stimulation system **400** continues to deliver the electrical stimulation signal through the selected electrical configuration. However, if the threshold has been met, the method **2100** returns to block **2104** in which the electrical stimulation system **400** randomly selects a new set of one or more electrodes **107** of the electrode array **408** to function as electrical sources and one or more other electrodes **107** of the electrode array **408** to function as electrical sinks. In other words, the electrical stimulation system **400** periodically updates the randomly selected set of source electrode(s) **107** and sink electrode(s) **107** through which the electrical stimulation signal is transmitted, for example, in order to ensure that the signal is delivered to the patient's tissue in a spatially random way.

[0125] Although the blocks **2102-2108** are described in a relatively serial manner, it should be appreciated that various blocks of the method **2100** may be performed in parallel in some embodiments.

[0126] Referring now to FIG. **22**, in use, the electrical stimulation system **400** may execute a method **2200** for providing spatially random electrical stimulation to a patient. It should be appreciated that the particular blocks of the method **2200** are illustrated by way of example, and such blocks may be combined or divided, added or removed, and/or reordered in whole or in part depending on the particular embodiment, unless stated to the contrary. It should be further appreciated that the method **2200** of FIG. **22** may be executed in conjunction with (e.g., in parallel with) the method **2100** of FIG. **21** described above.

[0127] The illustrative method **2200** begins with block **2202** in which the electrical stimulation system **400** (e.g., the controller **110**) partitions a frequency range into discrete frequency bands. For example, in an embodiment, the frequencies range may be from 0 to 100,000 Hz in equally spaced linear frequency bands (e.g., 1,000 Hz, 2,500 Hz, etc.). By way of example, FIG. **23** depicts a frequency range of a periodic signal that is partitioned into discrete frequency bands that have

1,000 Hz bandwidths. In another example, FIG. 24 depicts a broad spectrum electrical noise signal that has been partitioned/divided into **31** discrete tunable frequency bandwidths. It should be appreciated that the size of the bandwidths may differ depending on the particular embodiment. Further, although each of the bandwidths are described herein and depicted in FIGS. 23 and 24 as having the same width, it should be appreciated that one or more of the partitioned frequency bandwidths may have a different width from others in other embodiments. In an embodiment, each of the frequency bands in a frequency range of 0 Hz to 100 kHz of the partitioned spectrum has a bandwidth of one of 1 kHz or 2 kHz. Accordingly, it should be appreciated that the size of the bandwidths may differ depending on the particular embodiment. Additionally, the bandwidths may have the same or different widths depending on the particular embodiment. In some embodiments, the frequency may be described on a logarithmic scale and described by base2 (e.g., octaves: 10 Hz, 20 Hz, 40 Hz, 80 Hz, 160 Hz, 320 Hz, 640 Hz, 1280 Hz, 2560 Hz, etc.) and/or base-10 (e.g., 10 Hz, 100 Hz, 1,000 Hz, 10,000 Hz, 100,000 Hz, etc.) axis, with bandwidths described a Q-factor (center frequency/bandwidth) or bandwidth: octaves (1/10 to 10 oct).

[0128] In block **2204**, the electrical stimulation system **400** (e.g., the signal generator **109**) delivers the electrical stimulation signal through the randomly selected electrodes **107** of the electrode array **408** using the randomly selected electrical configuration of electrical sources and electrical sinks as described above.

[0129] In block **2206**, the electrical stimulation system **400** and/or the clinician receives feedback regarding the treatment from the patient. As described above, it should be appreciated that the feedback can be provided by the patient, for example, based on self-report (e.g., introspection, observations of unwanted sensory-and/or motor activity, autonomic dysfunction, etc.), determined from data generated by sensors measuring physiological outcomes, based on data from artificial intelligence or machine learning systems (e.g., output or feedback data), or based on combinations thereof. In some embodiments, the electrical stimulation signal(s) can be adjusted based on patient feedback describing unwanted sensations. Examples of unwanted sensations include pain and other sensations, such as urinary urgency and fecal urgency. Other examples include unwanted sensations of touch, vibration, pressure, tightness, feelings of warmth or cold, numbness, sounds of ringing (or buzzing, hissing, clicking, roaring, humming), anxiety, dizziness, aura, loss of sound intensity, and/or loss of sound quality. Any of the unwanted sensations described above can occur in the absence of an external stimulus and/or when the patient is perfectly still. Further, in some embodiments, the electrical stimulation signal(s) can be adjusted based on patient feedback describing physiological outcomes. It should be appreciated that the physiological outcomes may be reported by the patient, observed by a clinician, evaluated based on data generated by one or more sensors, and/or otherwise determined depending on the particular embodiment. Examples of physiological outcomes that can be used to adjust the electrical signals include those pertaining to bladder and bowel function, such as incontinence, constipation, voided volumes, voiding pressures, voiding frequency, and electromyogram signals of the urinary and colonic ensemble. Other examples of physiological outcomes that can be used to adjust the electrical signals include those pertaining to motor function, such as electromyogram signals, strength, weakness, stiffness, spasm, contracture, tremors, spasticity, atrophy, bradykinesia, and paralysis. Other markers of physiological activity that could be used to adjust the electrical signals include electroencephalogram signals, evoked brain potentials, and cognitive changes including dementia, hallucination, and delusion. Markers of autonomic function can also be used to adjust the electrical signals, such as heart-rate, blood flow, blood pressure, respiration, nausea, sweat production, skin color, and edema. Any of the physiological changes described above can be used to adjust the electrical signals in the absence of an external stimulus or when the patient is perfectly still. In some embodiments, sensor types include biopotential electrodes, pressure sensors and flow meters, chemical sensors, and/or temperature sensors (e.g., thermistors). It should be appreciated that, in some embodiments, one or more of the sensors can act in a close-loop fashion to adjust the

electrical signals to provide and optimize therapy.

[0130] In block **2206**, the electrical stimulation system **400** and/or the clinician determines whether to further adjust the signal applied to the patient based on the patient feedback (e.g., in an effort to most optimally alleviate the patient's condition). If not, the method **2200** returns to block **2204** in which the electrical stimulation system **400** (e.g., the signal generator **109**) delivers the electrical stimulation signal through the randomly selected electrodes **107** of the electrode array **408** using the randomly selected electrical configuration of electrical sources and electrical sinks. However, it should be appreciated that, in some circumstances, the particular randomly selected electrical configuration of electrical sources and electrical sinks may have changed relative to the prior execution of block **2204** due to the execution of the method **2100** of FIG. **21** described above in conjunction with the execution of the method **2200** of FIG. **22**.

[0131] However, if in block **2206**, the electrical stimulation system **400** and/or the clinician determines to adjust the signal applied to the patient based on the patient feedback (e.g., in an effort to most optimally alleviate the patient's condition), the method **2200** advances to block **2210** in which the electrical stimulation system **400** selects a particular frequency band of the discretely partitioned frequency bands for adjustment. In some embodiments, selection of the particular frequency band may involve selecting a corresponding signal generator **109** (in the case of multiple signal generators **109**), for example, in an embodiment in which each frequency band is powered by a different signal generator **109**. In block **2212**, the electrical stimulation system **400** adjusts the amplitude of the voltage and/or current of the signal within the selected frequency band, for example, by amplifying the voltage and/or current of the signal within that particular frequency band or attenuating the voltage and/or current of the signal within that particular frequency band. It should be appreciated that the amount of amplification/attenuation of the signal may be user-controlled, predefined, and/or otherwise determined depending on the particular embodiment. It should be appreciated that, in the illustrative embodiment, the clinician and/or patient may select and/or adjust the frequency band using one or more user interface **112** devices. However, the frequency band may be otherwise selected in other embodiments. In other embodiments, the electrical stimulation system **400** may additionally or alternatively adjust the phase and/or frequency of the periodic waveform corresponding with the selected frequency band in order to tune the patient.

[0132] The method **2200** returns to block **2204** in which the electrical stimulation system **400** (e.g., the signal generator **109**) delivers the electrical stimulation signal through the randomly selected electrodes **107** of the electrode array **408** using the randomly selected electrical configuration of electrical sources and electrical sinks. As indicated above, it should be appreciated that, in some circumstances, the particular randomly selected electrical configuration of electrical sources and electrical sinks may have changed relative to the prior execution of block **2204** due to the execution of the method **2100** of FIG. **21** described above in conjunction with the execution of the method **2200** of FIG. **22**. It should be further appreciated that the method **2200** allows for the patient and/or clinician to independently adjust the amplitude of the voltage and/or current of the electrical stimulation signal within multiple frequency bands (e.g., each frequency band) of that signal, such that the electrical stimulation signal is adjusted for optimal (or improved) treatment of the patient's condition. In some embodiments, the adjusted signal may be delivered to the patient continuously or intermittently (e.g., with durations of less than thirty minutes).

[0133] Although the blocks **2202-2212** are described in a relatively serial manner, it should be appreciated that various blocks of the method **2200** may be performed in parallel in some embodiments.

Claims

- 1.** A method of delivering spatially random electrical stimulation to a patient through an electrical lead that includes an electrode array, wherein the electrode array includes a plurality of electrodes, the method comprising: randomly selecting, by an electrical stimulation system, an electrical configuration from the plurality of electrodes of the electrode array, wherein randomly selecting the electrical configuration comprises randomly selecting a first set of electrodes of the plurality of electrodes to function as electrical sources and a second set of electrodes of the plurality of electrodes, different from the first set of electrodes, to function as electrical sinks; and delivering, by the electrical stimulation system, an electrical stimulation signal to the patient via the randomly selected electrical configuration.
- 2.** The method of claim 1, further comprising: partitioning, by the electrical stimulation system, a frequency range of the electrical stimulation signal into a plurality of discrete frequency bands, each of the discrete frequency bands having a corresponding bandwidth; and adjusting, by the electrical stimulation system, an amplitude of one or more of a voltage or a current of the electrical stimulation signal within a selected frequency band of the plurality of discrete frequency bands to generate an adjusted electrical signal based on feedback received from the patient; wherein delivering the electrical stimulation signal to the patient comprises delivering the adjusted electrical signal to the patient via the randomly selected electrical configuration.
- 3.** The method of claim 2, wherein the feedback comprises feedback received from the patient as patient self-report regarding the therapy delivered to the patient.
- 4.** The method of claim 2, wherein the feedback comprises feedback based on data generated by one or more sensors of the electrical stimulation system, wherein the one or more sensors measure one or more physiological outcomes of the patient.
- 5.** The method of claim 2, wherein the feedback comprises feedback data received from a machine learning system.
- 6.** The method of claim 2, wherein the feedback comprises a tissue impedance of the patient.
- 7.** The method of claim 1, further comprising randomly selecting, by the electrical stimulation system, a new electrical configuration from the plurality of electrodes of the electrode array in response to a determination that a threshold has been met.
- 8.** The method of claim 7, wherein the threshold is defined as a predefined period of time elapsed since randomly selecting the electrical configuration.
- 9.** The method of claim 7, wherein the threshold is associated with one or more characteristics of the spatially random electrical stimulation being delivered to the patient.
- 10.** The method of claim 1, wherein delivering the electrical stimulation signal to the patient via the randomly selected electrical configuration comprises directing a set of frequency components of the electrical stimulation signal to pass through a subset of the first set of electrodes and the second set of electrodes.
- 11.** The method of claim 1, wherein the electrical stimulation signal comprises a pulse waveform.
- 12.** The method of claim 1, wherein the electrical stimulation signal comprises a periodic waveform.
- 13.** The method of claim 1, wherein each electrode of the plurality of electrodes is cylindrical.
- 14.** The method of claim 1, wherein each electrode of the plurality of electrodes is spatially separated from each other electrode of the plurality of electrodes down a length of the electrical lead.
- 15.** The method of claim 1, wherein the plurality of electrodes comprises eight electrodes.
- 16.** The method of claim 1, wherein the first set of electrodes comprises a first number of electrodes and the second set of electrodes comprises a second number of electrodes different from the first number.
- 17.** The method of claim 1, wherein the electrode array comprises a distal section and a proximal section; and wherein delivering the electrical stimulation signal to the patient via the randomly

selected electrical configuration comprises delivering high-frequency signal components of the electrical stimulation signal through only the distal section of the electrode array.

18. The method of claim 1, wherein the electrode array comprises a distal section and a proximal section; and wherein delivering the electrical stimulation signal to the patient via the randomly selected electrical configuration comprises amplifying signals transmitted through the distal section at a different weight from signals transmitted through the proximal section.

19. The method of claim 1, wherein randomly selecting the first set of electrodes comprises randomly selecting a number of electrodes in the first set of electrodes; and wherein randomly selecting the second set of electrodes comprises randomly selecting a number of electrodes in the second set of electrodes.

20. The method of claim 1, wherein the frequency range of the electrical stimulation signal is 0.05 Hz to 2 kHz to block an action potential from firing.
